Can the Way Executives Speak Predict Stock Volatility?

Advanced NLP, Spring 2026, Assignment 3 – one-page summary

1. The question

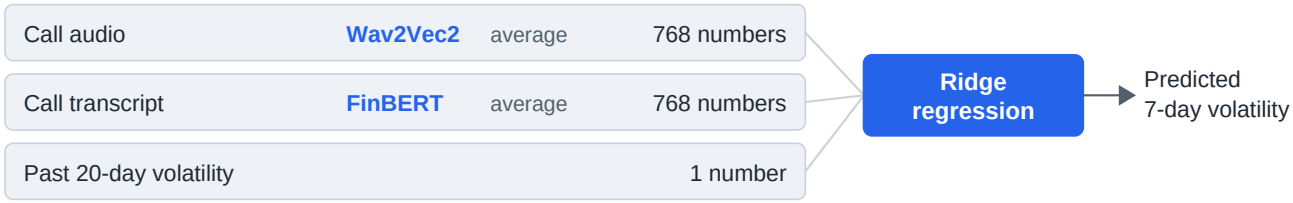
After a company reports earnings, its executives hold a conference call. We asked: **does the audio of that call (how things are said) help predict how much the stock will move in the following days, beyond the transcript (what is said) and the stock's own recent volatility?**

2. The dataset

- **Earnings calls:** Qin & Yang (ACL 2019) dataset, 572 S&P 500 calls from 2017 (280 companies). Each call has executive speech split into sentences, with the text and the matching audio clip (89,722 sentences in total).
- **Stock prices:** daily prices downloaded from Yahoo Finance. 490 calls (240 companies) had prices; 82 calls from 40 delisted companies had none and were left out.
- **What we predict:** how much the stock price swings over the 7 trading days after the call (the log of the standard deviation of daily returns), starting the day after the call.

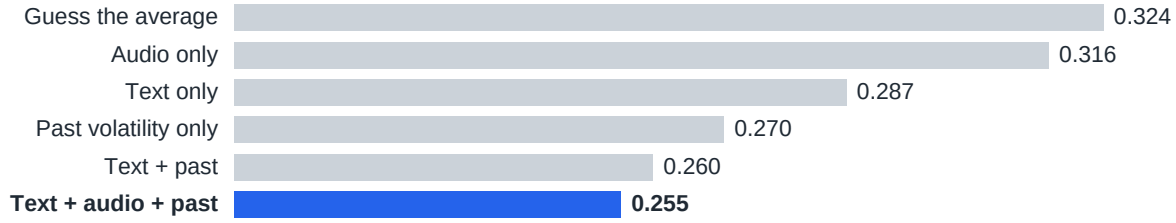
3. The experiment

Each call is turned into numbers using two pre-trained models that we did not retrain: **FinBERT** (a language model for financial text) and **Wav2Vec2** (a speech model). A simple **ridge regression** (linear regression that keeps weights small to avoid overfitting) then makes the prediction.



We split by time: **train** on January–June calls (251), **tune** on July–September (123), and **test** on October–December (110). We compared every combination of inputs, using mean squared error (MSE) on the test calls, where lower is better.

4. Results (test error, lower is better)



Bars start at 0.20 to make the differences visible.

5. Observations

- **The stock's own recent volatility is the strongest clue.** Every model without it did worse than using it alone.
- **Adding text and audio together gave the best model.** It beat past volatility alone, and this gain is statistically significant.
- **Audio's own contribution is small.** On top of text and past volatility, it improved error by only 0.005. That is in the right direction but not statistically significant. Audio by itself had almost no predictive signal.
- **The gains do not carry over to new companies.** When tested on companies never seen in training, past volatility alone was best. The models may partly be recognising the company or speaker rather than learning from the content or tone.
- **FinBERT was not clearly better than simple word counts (TF-IDF).**

6. Bottom line

Text and speech together improve volatility forecasts slightly, but we cannot confirm that speech alone adds value. The test set is small (110 calls), so a larger dataset and models that use sentence-by-sentence alignment would be the next step.